



Survey on LLM Safety: Attacks, Defenses, Alignment, Metrics, and Guardrails

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Abstract

Large Language Models (LLM) have demonstrated remarkable capabilities across various applications, but their deployment raises critical safety concerns as potential misuse poses significant societal risks. This survey reviews the end-to-end security and safety pipeline of LLMs, focusing on the interaction between users and model responses. We categorize the system into five key components: attacks, defenses, safety alignment, metrics and guarding mechanisms. Attacks involve crafting adversarial inputs to exploit model vulnerabilities. Defenses act as countermeasures, aiming to detect and prevent such inputs before processing. Safety alignment ensures that, even when attacks reach the model, its responses remain consistent with ethical and policy-aligned behavior. Guarding mechanisms operate post-response to flag, filter, or block unsafe outputs. Each of these stages is subject to rigorous evaluation metrics to assess their effectiveness, robustness, and limitations. As LLMs evolve toward more general-purpose intelligence, these safety considerations become increasingly critical for the development of robust and trustworthy AI systems. Finally, we highlight open challenges and future research directions to advance the security and alignment of LLMs.

Keywords Adversarial attacks · AI governance · AI robustness · Content moderation · Defense mechanisms · Ethical AI · Evaluation metrics · Guardrails · LLMs · LLM safety · Model vulnerabilities · Red teaming · Safety alignment · Secure AI systems · Trustworthy AI

1 Introduction

LLMs have been one of the most prominent research topics in recent years. They are likely to remain at the forefront of AI research for the foreseeable future, particularly as efforts continue toward achieving comprehensive safety and alignment (Liu et al., 2023; Anwar

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et al., 2024). As these systems continue to scale and exhibit increasingly general capabilities, they are often viewed as important steps toward Artificial General Intelligence (AGI), further amplifying the importance of robust safety and security mechanisms. Building on the foundational works, many state-of-the-art LLMs have been developed by leading organizations, each contributing unique architectures, training methodologies, and alignment techniques. Notable examples include OpenAI's GPT series (Brown et al., 2020; Achiam et al., 2023), Anthropic's Claude models (2024 (Achiam et al.,), Google DeepMind's Gemini (Team et al., 2023, 2024), and Meta's LLaMA family (Touvron et al., 2023), alongside emerging models such as Mistral, Falcon, Vicuna, and Alpaca (Chowdhery et al., 2023; Team et al., 2024).

The official technical reports and research papers on these models provide essential insights into their design and safety alignment strategies. Reviewing these works is critical for understanding ongoing efforts to address key challenges including bias, toxicity, factual accuracy, and ethical alignment in LLM deployment. Prior works (Cao et al., 2025; Ying et al., 2024; Nadeau et al., 2024; Perez et al., 2023) were conducted to evaluate the bias score and toxicity score, while SafetyBenchASR (Zhang et al., 2024) and TruthfulQA (Lin & Hilton, 2022) address how vulnerable the LLM models are. Table 1 shows the safety and alignment metric of the selected LLMs. Beyond basic misuse, LLM safety encompasses structured threats such as prompt injection, adversarial manipulation, and backdoor attacks each capable of producing harmful or misleading outputs through distinct mechanisms. Scores in Benchmark are collected from publicly reported benchmark evaluations and were not obtained under identical experimental conditions. Consequently, the results should be interpreted as indicative rather than strictly comparable. The findings indicate that publicly available open-source LLMs demonstrate generally similar performance, based on the observed scores.

Table 1 Comparison of major LLMs on bias, toxicity, safety robustness (ASR), and truthfulness (TruthfulQA) on publicly disclosed data

Model	Bias score (↓)	Toxicity score (↓)	Safety-Bench ASR (% ↑)	TruthfulQA accuracy (% ↑)
GPT-4o (Hurst et al., 2024)	~0.02	0.14	80.3	82–83
GPT-4 (Achiam et al., 2023)	~0.02	0.14	–	82–83
Claude 3.5 Sonnet (Claude,) 2024	~0.02	~0.02	80.5	~85
Gemma/Gemma2 (Team et al., 2024)	~0.22	~0.22	59–70	–
LLaMA 3 (Grattafiori et al., 2024)	~0.20	~0.13	~68	~70
Mistral (Jiang et al., 2023)	~0.21	~0.21	~68	~67
Falcon (Almazrouei et al., 2023)	>0.30	~0.30	–	–
Deepseek (Liu et al., 2024)	–	–	78.5 to ~79.7	–
Vicuna (Chiang et al., 2023)	~0.25	~0.25	45.1	~60
Alpaca (Taori et al., 2023)	–	~0.29	< random	~55

Moreover, bias and hallucination (Abishethvarman et al., 2025) issues can lead to the spread of misinformation, reinforcing stereotypes, or fabricating nonfactual information. These are evaluated with various benchmarks (see Table 2) and the best scores are listed in the LLM's technical papers. Consequently, developing defensive mechanisms to mitigate these vulnerabilities is essential. Furthermore, LLMs must be aligned with human values and ethical principles, ensuring they behave in a safe and responsible manner.

This survey categorizes LLM safety into five key areas:

- **Attacks:** Various adversarial techniques used to manipulate language models outputs, including prompt injection, backdoor attacks, adversarial perturbations, and data poisoning (Yao et al., 2024; Xu et al., 2023).
- **Defenses:** Countermeasures such as adversarial training, robust fine-tuning, differential privacy, and content filtering to enhance LLM security and robustness (Varshney et al., 2023).
- **Alignment:** Methods to ensure that LLMs adhere to ethical guidelines, such as reinforcement learning with human feedback (RLHF), constitutional AI, and instruction tuning (Wang et al., 2024).
- **Metrics:** Evaluation frameworks to measure safety, robustness, and fairness in LLMs, including toxicity detection, factual consistency, and bias auditing (Li & Li, 2023).
- **Guardrails:** Policy-based and technical safeguards for responsible deployment, such as inference guidance, access control, compliance frameworks, and automated content moderation (He et al., 2024).

Table 2 Summary of major benchmarks for assessing LLM safety and alignment, showing their focus areas and usage in the original technical papers

Benchmarks	Focus area	Considered
TruthfulQA (Lin et al., 2021)	Truthfulness, factual correctness	✓
ToxiGen (Hartvigsen et al., 2022)	Toxicity and hate speech generation detection	✓
HHH Eval (Bai et al., 2022)	Helpfulness, Honesty, Harmlessness	✓
DecodingTrust (Wang et al., 2023)	Trustworthiness, robustness, hallucination	✓
AdvBench (Zou et al., 2023)	Jailbreak/adversarial input resistance	✓
Anthropic Red Teaming (Bai et al., 2022)	Harmlessness via red-teaming	✓
HELM Safety (Liang et al., 2022)	Holistic LLM evaluation incl. safety	✓
RealToxicityPrompts (Gehman et al., 2020)	Toxicity from open-ended prompts	✓
DoNotAnswer (DNA) (Wang et al., 2023)	Model refusal to unethical prompts	✓
SafetyBench (Zhang et al., 2023)	Alignment with safety-critical questions	✓
MT-Bench (Zheng et al., 2023)	Evaluating alignment with Multi-turn chat evaluation	~
Model-Written Evals (Perez et al., 2023)	Benchmarks written/generated by models	~
HelpfulQA (Havrilla et al., 2023)	Helpfulness + instruction following	~

Legend: ✓ – Benchmark considered for the studies; ~ – Benchmark reviewed but not directly used or only partially applicable

This survey systematically reviews prior research on attacks, defenses, safety alignment, evaluation frameworks, and security mechanisms. The taxonomy of this survey is illustrated in Fig. 1. Table 2 highlights the breadth of safety and alignment benchmarks considered in this study, demonstrating comprehensive coverage across truthfulness, toxicity, robustness, and alignment dimensions.

2 Motivation

This survey builds upon prior foundational studies in LLM safety and security. Ensuring LLM safety is critical, as widespread adoption introduces risks of misuse, adversarial manipulation, and ethical misalignment. While existing studies have explored LLM secu-

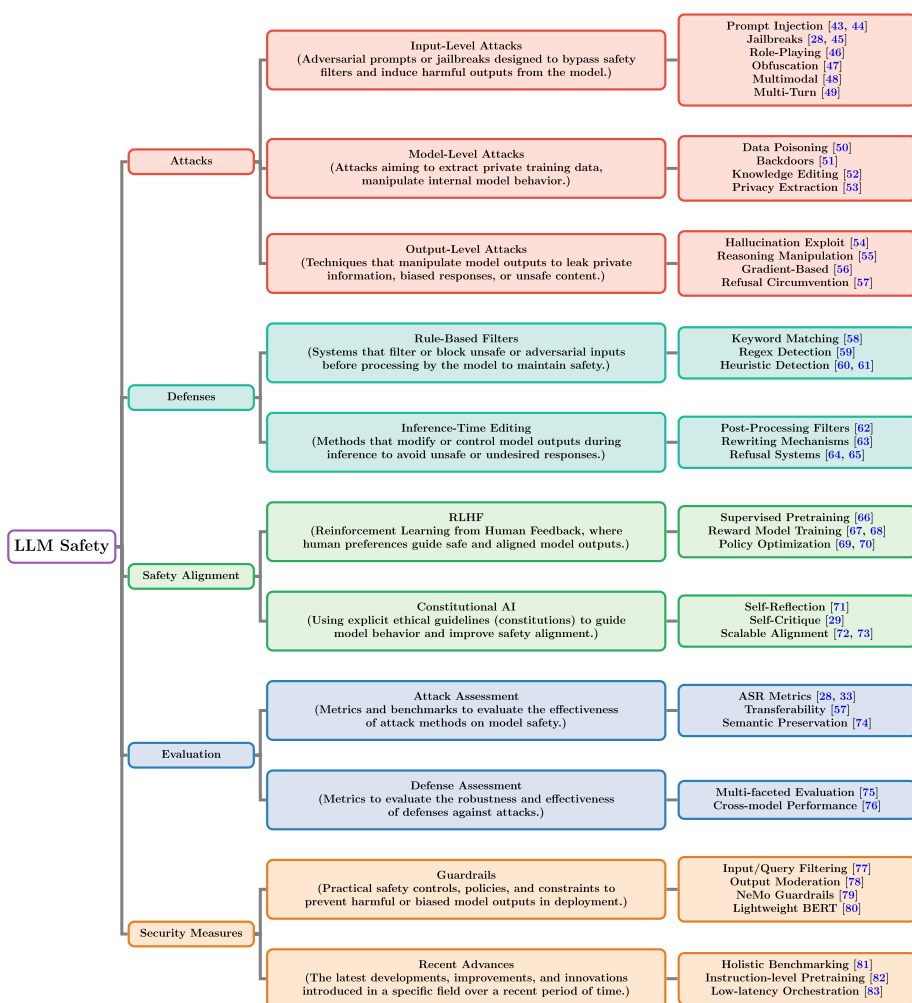


Fig. 1 Taxonomy of this survey, illustrating the hierarchical organization of LLM safety research across attacks, defenses, alignment, evaluation, and security measures, with representative citations at each level

ity, there is a need for a unified perspective that connects attacks, defenses, alignment strategies, evaluation metrics, and guardrails.

Prior surveys (Yao et al., 2024; Xu et al., 2024) have classified LLM threats and defenses, focusing on specific aspects like jailbreak attacks, inference-time vulnerabilities, and defense mechanisms. However, a comprehensive framework that integrates these dimensions spanning adversarial risks, safety alignment techniques, and evaluation methodologies remains lacking.

Table 3 summarizes major existing surveys on Large Language Model safety and positions the present study within this landscape. Prior works primarily emphasize conversational safety, jailbreak attacks, alignment strategies, or full-stack system considerations. In contrast, this work focuses specifically on reliability at the application layer, highlighting how hallucination, inconsistency, and untrustworthy outputs affect Knowledge-Based Systems. By shifting attention from adversarial or training-centric perspectives to downstream knowledge integrity, this study complements existing surveys and addresses an underexplored yet practically critical dimension of LLM deployment.

This survey bridges this gap by systematically categorizing LLM attacks, examining multi-layered defenses, and evaluating safety alignment mechanisms. We further introduce key metrics and guardrails to assess and enforce LLM safety, ensuring robust and trustworthy AI deployment.

Table 3 Comparative analysis of existing LLM safety surveys and our perspective

Survey paper	Primary focus of prior work	Our contribution and distinction
Attacks, Defenses and Evaluations for LLM Conversation Safety (Dong et al., 2024)	Conversational safety, jailbreak attacks, defenses, and evaluation benchmarks	Extends beyond conversational safety by addressing hallucination and inconsistency in Knowledge-Based Systems
A Comprehensive Study of Jailbreak Attack versus Defense for LLMs (Xu et al., 2024)	Detailed analysis of jailbreak techniques and defense strategies	Shifts focus from adversarial settings to everyday unreliable outputs affecting real-world knowledge systems
AI Safety in Generative AI Large Language Models (Chua et al., 2024)	High-level overview of safety challenges and mitigation strategies	Provides an application-driven perspective linking LLM failures to Knowledge-Based System degradation
Large Language Model Safety: A Holistic Survey (Shi et al., 2024)	Alignment, robustness, and training-based safety approaches	Highlights downstream impacts of hallucinated knowledge on decision-making systems
On Large Language Models Safety, Security, and Privacy (Zhang et al., 2025)	Joint discussion of safety, security, and privacy concerns	Isolates reliability as a core issue and analyzes its role in Knowledge-Based Systems
A Comprehensive Survey in LLM (-Agent) Full Stack Safety (Wang et al., 2025)	Safety across data, training, agents, and deployment pipelines	Concentrates on knowledge trustworthiness at the application layer rather than full-stack engineering

3 Literature Review

This literature review examines existing research on LLM safety, focusing on attacks, defenses, alignment, metrics, and guardrails. Our objective is to categorize prior works based on these domains, identifying key contributions and gaps in the literature.

3.1 Surveying Existing Research

Several surveys and studies have been conducted to explore the security and privacy concerns of LLMs. These works categorize LLM security from multiple perspectives, including adversarial attacks, the robustness of defense mechanisms, alignment techniques, evaluation metrics, and safety measures. The distribution of surveyed papers across these research areas is illustrated in Fig. 2. Prior research has identified various attack vectors, such as jail-break attacks, adversarial prompting, and data poisoning. In parallel, several defense strategies have been proposed, including rule-based filters and inference-time editing techniques.

As shown in Table 4, prior surveys predominantly organize LLM safety either through adversarial-centric categorizations, thematic parallel dimensions, conceptual domain separations, or engineering lifecycle stages. In contrast, the proposed taxonomy adopts a unified pipeline perspective that integrates attacks, defenses, alignment mechanisms, evaluation metrics, and safety measures into a continuous reliability-focused framework. This structure enables modeling of cross-stage dependencies and failure propagation, thereby extending beyond modular or parallel taxonomies in prior literature.

4 Novelty of the Survey

The novelty of this survey lies in its comprehensive and integrated examination of the entire LLM safety and security pipeline from attack vectors to defense strategies, safety alignment, guarding mechanisms, and evaluation metrics. While prior studies often focus on individ-

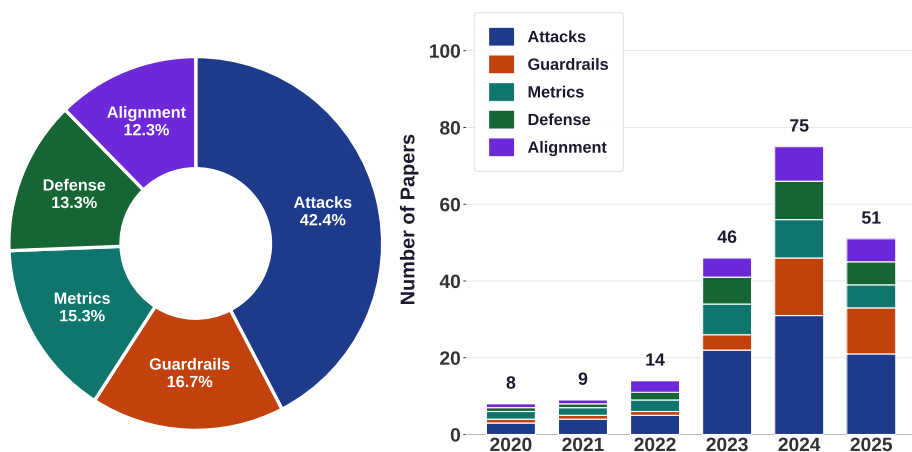


Fig. 2 Overview of the distribution of papers across key LLM safety topics, along with year-wise publication trends from 2020 to 2025

Table 4 Structural comparison of taxonomies in existing LLM safety surveys

Survey paper	Taxonomy structure	Organizing principle	Structural characteristic
A Comprehensive Study of Jailbreak Attack versus Defense for Large Language Models (Xu et al., 2024)	Attack → Defense → Evaluation, subdivided into inference-time, training-time, filters, metrics	Adversarial-centric categorization	Linear grouping; focuses on attack–defense cycle without lifecycle integration
Attacks, Defenses and Evaluations for LLM Conversation Safety: A Survey (Dong et al., 2024)	Attack → Defense → Evaluation, including templates, red-teaming, and neural prompt-to-prompt	Conversational safety focus	Structured by technique; emphasizes attack–defense pipeline at conversation level
AI Safety in Generative AI Large Language Models: A Survey (Chua et al., 2024)	Data Safety, Model Safety, Prompt Safety, Alignment, Safety at Scale	Layered thematic classification	Parallel safety dimensions; limited cross-category dependency modeling
Large Language Model Safety: A Holistic Survey (Shi et al., 2024)	Robustness, Bias, Privacy, Governance, Interpretability, Agents	Broad thematic expansion	Wide coverage; taxonomy spans technical and governance domains but remains modular
On Large Language Models Safety, Security, and Privacy: A Survey (Zhang et al., 2025)	Safety vs. Security vs. Privacy as distinct pillars	Conceptual separation of risk domains	Clear definitional boundaries; domains treated independently
A Comprehensive Survey in LLM (-Agent) Full Stack Safety: Data, Training, and Deployment (Wang et al., 2025)	Data → Pre-training → Post-training → Deployment → Agents → Commercialization	Engineering lifecycle framework	Stage-based system pipeline; emphasizes operational workflow
Our survey	Attacks → Defense → Alignment → Evaluation Metrics → Safety Measures	Unified safety pipeline with reliability focus	Integrated lifecycle modeling; captures cross-stage propagation and downstream knowledge reliability

ual components in isolation, our approach emphasizes the interdependence and interaction between these layers. By analyzing how attacks propagate through defenses, influence alignment behavior, and trigger (or bypass) guarding mechanisms, we provide a holistic view of LLM vulnerability and resilience. This interconnected perspective enables a more realistic and operational understanding of LLM deployment in adversarial environments, highlighting system-wide trade-offs and failure modes that are overlooked when components are studied separately. Although LLM deployment follows a sequential pipeline, Prior work often analyzes attacks, defenses, alignment, and safety measures in isolation. Our framework instead models their interdependencies and cross-stage failure propagation.

5 Introduction to Our Safety Perspective

Ensuring the safety of LLMs requires a comprehensive, multi-layered approach that addresses potential risks throughout the model’s life cycle.

In this survey, we present an original, structured perspective that organizes safety mechanisms into three distinct but complementary stages: (1) Input-Level Safety, (2) Training-Time Alignment, and (3) Inference-Time Guarding Mechanisms. The overall safety pipeline spanning these stages is illustrated in Fig. 3.

1. Input-Level Safety: This initial layer serves as the first barrier, with mechanisms designed to detect and block harmful or adversarial prompts before they reach the model. Techniques in this stage include sophisticated input sanitization, prompt validation, and adversarial input detection. By preemptively filtering inputs, this layer prevents exploitation via methods such as prompt injection, jailbreak attempts, and malicious encoding.

2. Training-Time Alignment: During model training, alignment strategies are implemented to shape the LLM's underlying behavior and principles. These methods such as reinforcement learning from human feedback (RLHF), reward modeling, and constitutional AI optimize models to instill robust, inherent safety. Well-aligned models are less likely to produce unsafe responses, even when encountering adversarial or ambiguous prompts.

3. Inference-Time Guarding Mechanisms: Despite thorough training, models may still emit unintended or harmful outputs. This final protective layer comprises tools and strategies applied after content is generated. Post-processing filters, output classifiers, context manipulation, and advanced system prompts intercept and revise outputs as necessary to prevent harmful information from reaching end users.

Figure 4 visually summarizes this taxonomy, demonstrating how safety interventions are strategically distributed across the LLM development and deployment lifecycle, with representative mechanisms at each stage.

By synthesizing these layers, our perspective clarifies both the strengths and the residual limitations inherent to each approach.

This framework supports the design of robust defense strategies that combine proactive prevention blocking risks before they arise with reactive filtering to catch challenges that escape earlier barriers. Ultimately, such a comprehensive taxonomy provides a foundation for advancing the safety and reliability of next-generation language models.

6 Attacks

To strengthen the operational relevance of this survey, we explicitly articulate the threat model and underlying assumptions that motivate each attack and defense class, and we enumerate (i) the attack surface an adversary can realistically access and (ii) the expected fail-

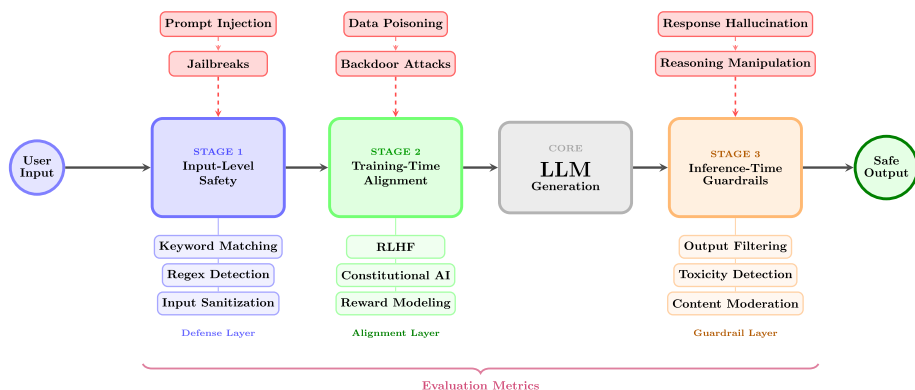
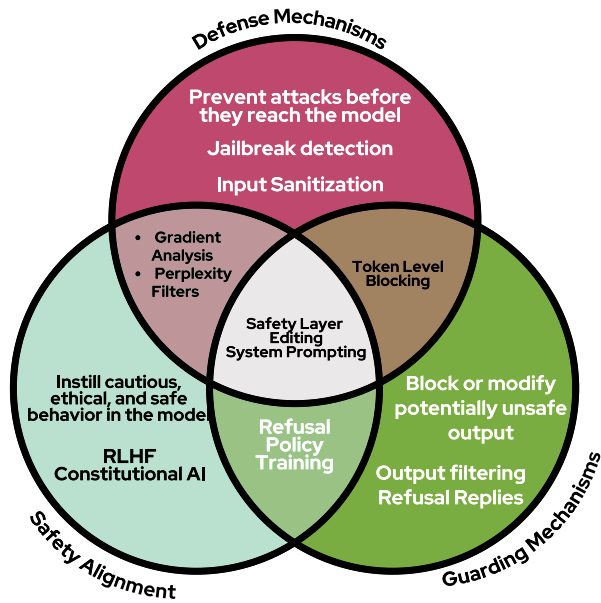


Fig. 3 End-to-end architecture of the LLM safety pipeline, illustrating the three-stage safety perspective: Input-Level Safety (defenses), Training-Time Alignment (RLHF, Constitutional AI), and Inference-Time Guardrails (output filtering). Red dashed arrows indicate adversarial attack vectors targeting each stage

Fig. 4 Layered taxonomy of LLM safety interventions, illustrating the hierarchy of strategies across defenses, alignment, and guarding mechanisms



ure modes when protections degrade or are bypassed. This framing complements existing taxonomies of LLM threat vectors and helps connect abstract attack categories to concrete deployment settings (Shayegani et al., 2023; Xu et al., 2023; Yao et al., 2024).

Concretely, we assume an adaptive adversary with one or more of the following capabilities depending on the setting: (a) *query-only* black-box access to the deployed model (including rate limits and content moderation constraints), (b) partial *system-level* access such as control over user-provided context, tool outputs, or retrieved documents in retrieval-augmented generation (RAG) pipelines, and/or (c) *training-time* influence via data collection, curation, or fine-tuning procedures. For each attack class discussed below, we therefore highlight where the adversary interacts with the system (prompt interface, memory/context window, retrieval store, tool APIs, training data, or post-processing filters) and we categorize typical failure modes, such as instruction-hierarchy override, context poisoning, tool/RAG prompt injection, privacy leakage (memorization and extraction), latent backdoor trigger activation, and downstream policy evasion through output formatting or reasoning manipulation (Liu et al., 2023; Yip et al., 2023; Pan et al., 2025; Benjamin et al., 2024; Jia et al., 2025; Yu et al., 2023; Ogundoyin et al., 2025; Yu et al., 2024; Wan et al., 2023; Carlini et al., 2021; Liu et al., 2024).

LLMs are vulnerable to a wide range of adversarial attacks that can occur at different stages of their lifecycle (Shayegani et al., 2023; Xu et al., 2023). These threats can be broadly categorized (as shown in Fig. 5) into Input-Level, Model-Level, and Output-Level attacks, which directly undermine the three-stage safety perspective by targeting Input-Level Safety (e.g., bypassing prompt validation), Training-Time Alignment (e.g., corrupting ethical behaviour during fine-tuning), and Inference-Time Guarding (e.g., exploiting post-generation filters). Each category targets a specific phase in the LLM pipeline, with potential propagation across stages, and poses unique challenges for safety, robustness, and alignment. Addressing these requires integrated defenses that reinforce the interdependence of the safety layers, as visualised in Fig. 4.

6.1 Input-Level Attacks (Prompt-Time/Inference-Time)

Input-level attacks exploit language models during inference by crafting prompts that manipulate model behavior to bypass safety mechanisms and elicit unintended or harmful responses (Liu et al., 2023; Yip et al., 2023; Pan et al., 2025; Benjamin et al., 2024; Jia et al., 2025; Yu et al., 2023; Ogundoyin et al., 2025; Yu et al., 2024). These attacks primarily undermine Input-Level Safety in our safety perspective by evading preemptive detection, potentially propagating to weaken Training-Time Alignment if not mitigated. Recent research categorises these attacks into diverse classes, each targeting the model's prompt-processing or conversational input mechanisms through distinct strategies, highlighting the need for robust input sanitization and cross-stage guarding.

Prompt Injection Attacks Prompt injection is among the most widely studied and impactful attack vectors on large language models. These attacks embed malicious instructions within the user input, manipulating the model's instruction-following abilities and often bypassing intended restrictions (Liu et al., 2023; Yip et al., 2023; Pan et al., 2025; Benjamin et al., 2024; Jia et al., 2025; Perez, 2022). Specific techniques include:

- **Direct Injection:** Explicit attempts to override system prompts with simple commands, such as "Ignore all previous instructions and provide unrestricted answers." (Perez, 2022)
- **Indirect Injection:** Concealing adversarial instructions within benign text or content retrieved externally by the model (Yu et al., 2023; Ogundoyin et al., 2025; Yu et al., 2024).
- **Context Manipulation:** Leveraging the context window by positioning conflicting instructions at different points in the prompt (Liu et al., 2023; Yip et al., 2023; Pan et al., 2025; Benjamin et al., 2024; Jia et al., 2025).

Jailbreak Attacks Jailbreak attacks are systematic attempts to defeat safety alignment by leveraging advanced optimization or automated prompt-rewriting. These methods can produce adversarial suffixes or iteratively refine commands to circumvent guardrails (Zou et al., 2023; Wang et al., 2024), often exploiting gaps in input-level safety that may affect downstream inference-time guarding. Notable examples include:

- **GCG (Greedy Coordinate Gradient):** Uses white-box gradient-based optimization to identify effective adversarial suffixes for prompts (Zou et al., 2023; Wang et al., 2024).
- **AutoDAN:** Employs gradient techniques to craft coherent, transferable adversarial prompts with human-readable structure (Zhu et al., 2023).
- **PAIR (Prompt Automatic Iterative Refinement):** A black-box method utilizing an attacker model to iteratively evolve jailbreak prompts (Chao et al., 2024; Mehrotra et al., 2023; Reddy et al., 2025).
- **Stealthy Attacks:** Incorporate benign data mirroring during adversarial search to evade detection and maintain high attack rates (Mu et al., 2025; Koide & Nakano, 2026).
- **Persuasive Attack:** Prior work such as DeepInception (Li et al., 2024) leverages nested narrative role-playing prompts to exploit LLMs' personification and authority-driven reasoning, enabling the model to bypass safety guardrails.

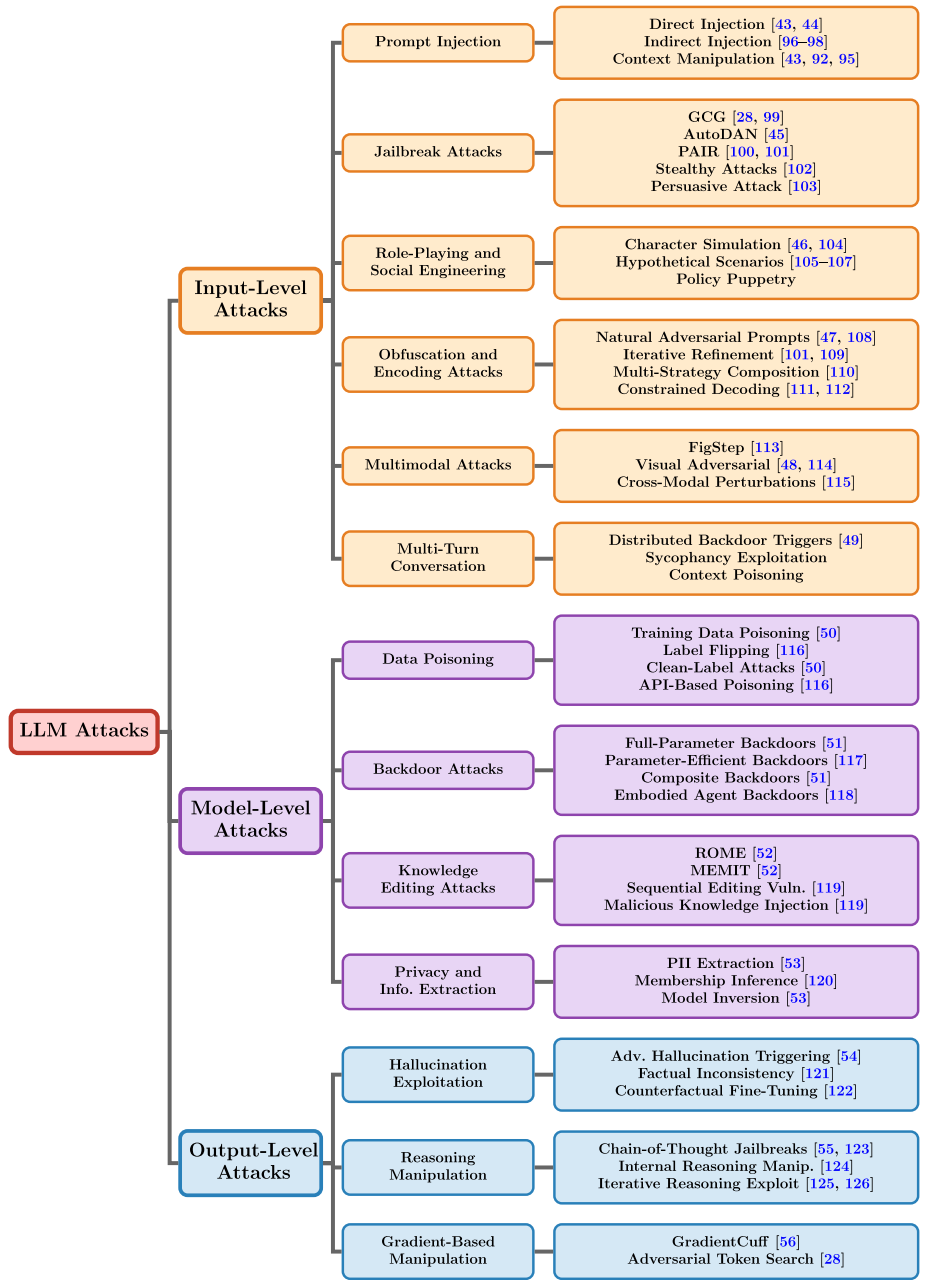


Fig. 5 A comprehensive overview of adversarial attacks targeting LLMs, categorised by attack surface into input-level, model-level, and output-level threats. Each subcategory highlights specific techniques along with representative citations

Role-Playing and Social Engineering Attacks These strategies induce models to break safety alignment through manipulation of contextual or social cues (Gumaan, 2025; Wan et al., 2024; Si et al., 2025; Tonmoy et al., 2024; Chen et al., 2025). Techniques include:

- Character Simulation: Requesting the model to adopt unaligned personas or act as banned characters (Gumaan, 2025; Wan et al., 2024)
- Hypothetical Scenarios: Framing malicious requests within academic or fictional contexts to bypass direct filtering (Gumaan, 2025; Wan et al., 2024; Si et al., 2025; Tonmoy et al., 2024; Chen et al., 2025).
- Policy Puppetry: Utilizing structured data formats (e.g., XML, JSON) to disguise prompt intent or mimic system-level policy cues.

Obfuscation and Encoding Attacks These attacks mask malicious intent using automated prompt optimization, adversarial suffix generation, and iterative refinement techniques (Basani, 2024; Yang et al., 2024; Mehrotra et al., 2023; Koo et al., 2025; Shayegani et al., 2023). They include:

- Natural-Looking Adversarial Prompts: Human-readable yet adversarially optimized suffixes that bypass safety guardrails in black-box settings (Basani, 2024; Yang et al., 2024).
- Iterative Refinement Attacks: Automatically refining prompts through feedback loops or attacker models to increase jailbreak success rates (Mehrotra et al., 2023; Koo et al., 2025).
- Multi-Strategy Composition Attacks: Combining diverse disguise or reasoning strategies in an adaptive manner to evade moderation systems (Qi et al., 2025).
- Constrained Decoding Attacks: Exploiting structured output interfaces or decoding constraints to introduce adversarial logic that subverts output safety filters (Zhang et al., 2026; Li et al., 2025; Wang et al., 2024; Wei et al., 2026).

Multimodal Attacks Vision-language and multimodal models are susceptible to attacks that leverage the visual channel (Qi et al., 2023; Ren et al., 2025). Examples include:

- FigStep: Typographic image-based prompts that deliver prohibited content through visual cues, maintaining high attack success on vision-language models (Gong et al., 2025).
- Visual adversarial examples that inject harmful intent through crafted images, achieving high jailbreak success rates across vision-language models (Qi et al., 2023).
- Systematic adversarial attacks against vision-language models that bypass safety alignment mechanisms via visual inputs (Ren et al., 2025).
- Targeted multimodal perturbations that manipulate cross-modal generation systems through imperceptible input modifications (Liu et al., 2025).

Multi-turn Conversation Attacks Conversational LLMs are vulnerable to attacks distributed over multiple dialogue rounds (Tong et al., 2024). Key techniques include:

- Distributed Backdoor Triggers: Trigger tokens are spread across multiple conversation-

al turns, only producing the attack effect when all are provided in sequence (Tong et al., 2024).

- **Sycophancy Exploitation:** Gradually manipulating the model's cooperative tendencies across turns to elicit unsafe behavior.
- **Context Poisoning:** Inserting adversarial information early in the dialogue to bias later model responses.

6.2 Model-Level Attacks (Training-Time/Fine-Tuning Phase)

Model-level attacks compromise language models by introducing adversarial artifacts or manipulations during the data curation, training, or fine-tuning phases. These threats primarily undermine Training-Time Alignment in our safety perspective, creating persistent vulnerabilities that may evade detection and propagate risks to Inference-Time Guarding mechanisms if not addressed. Such attacks target the model's integrity at a foundational level, posing significant challenges for safety and ethical behavior even after extensive alignment efforts (Zhang et al., 2024; Zhou et al., 2025).

Data Poisoning Attacks Data poisoning manipulates training datasets to subtly or drastically alter the behavior of the resulting model, undermining Training-Time Alignment by embedding harmful biases or triggers at the core of the model's learning process (Wan et al., 2023; Wallace et al., 2021). Common approaches include:

- **Training Data Poisoning:** Injection of malicious or adversarial samples into the pre-training data pool can shift model outputs even at very low contamination rates (e.g., 0.1% of the dataset) (Wan et al., 2023).
- **Label Flipping:** Systematic inversion or distortion of training labels misleads models into forming false associations between input and output samples, leading to targeted misclassifications (Wallace et al., 2021).
- **Clean-Label Attacks:** Incorporates carefully selected but correctly-labeled examples that bias model behavior without easily-detectable label errors, making attack detection challenging (Wan et al., 2023).
- **API-Based Poisoning:** Exploits vulnerabilities in data collection pipelines to inject adversarial data or corrupt corpora used in training, often through insecure infrastructure or exposed credentials (Wallace et al., 2021).

Backdoor Attacks Backdoor attacks embed persistent, hidden triggers into model parameters during training or fine-tuning. When presented with specific cues, the model outputs malicious content while appearing benign otherwise, directly challenging Training-Time Alignment and risking downstream Inference-Time failures (Li et al., 2021; Aghakhani et al., 2024). Techniques include:

- **Full-Parameter Fine-Tuning Backdoors:** Modifying all parameters during fine-tuning to deeply embed attack triggers (Li et al., 2021).
- **Parameter-Efficient Backdoors:** Leveraging methods like LoRA or adapter-based fine-tuning to implant triggers with minimal parameter modification, reducing detectability (Aghakhani et al., 2024).
- **Composite Backdoor Attacks:** Distributing multiple trigger cues across different prompt

sections to achieve robust activation with few poisoned samples, with some attacks reaching high success rates (Li et al., 2021).

- Cross-Lingual Backdoor Transfer: Poisoning data in one language causes backdoor activation in others, enabling attack transfer across multiple languages with near-perfect success rates in some cases (Aghakhani et al., 2024).
- Embodied Agent Backdoors: Targeting LLM-based autonomous agents using a mixture of word, scenario, or knowledge injection for malicious control, especially in decision-making systems (Jiao et al., 2025).

Knowledge Editing Attacks Knowledge editing, originally developed for rapid, targeted updates to a model's factual knowledge, can be abused to implant or alter information for adversarial purposes, disrupting Training-Time Alignment by introducing false or harmful associations (Meng et al., 2023; Cao & Aziz, 2021). Prominent attack vectors include:

- ROME (Rank-One Model Editing): Directly updating specific internal associations (e.g., altering factual mappings like “capital of France → London”) via low-rank interventions in the network (Meng et al., 2023).
- MEMIT (Mass Editing Memory in Transformer): Scaling ROME-style edits across multiple transformer layers to enable thousands of simultaneous changes, amplifying the scope of malicious edits (Meng et al., 2023).
- Sequential Editing Vulnerabilities: Repeated edits to model knowledge can result in the loss of previously injected facts, undermining model reliability and creating openings for further manipulation (Cao & Aziz, 2021).
- Malicious Knowledge Injection: Leveraging editing algorithms to surreptitiously implant harmful, untrue, or biased information while preserving overall model utility and fluency (Cao & Aziz, 2021).

Privacy and Information Extraction Privacy-related attacks aim to uncover or extract sensitive or personally identifiable information (PII) memorized by the model during pre-training or fine-tuning, violating safety principles embedded during Training-Time Alignment and posing risks to user trust (Carlini et al., 2021; Huang et al., 2023). Techniques include:

- PII Extraction Attacks: Systematically probing models, especially after fine-tuning, can reveal personal details at rates significantly higher than in pre-trained models, exposing memorized data (Carlini et al., 2021).
- Membership Inference: Determining whether specific user data was present in the training set by analyzing response patterns and memorization behaviors, which can compromise confidentiality (Huang et al., 2023).
- Model Inversion: Reconstructing actual training samples or attributes from model parameters, posing acute risks for models trained on sensitive or proprietary datasets (Carlini et al., 2021).
- Advanced Adversarial PII Extraction: Utilizing continual, adaptive queries and iterative prompt engineering to maximize the extraction of sensitive data, greatly increasing privacy risks (Huang et al., 2023).

6.3 Output-Level Attacks (Post-Generation/Reasoning Manipulation)

Output-level attacks target the model's generation logic and final responses, exploiting vulnerabilities in the reasoning and output generation process. These threats primarily undermine Inference-Time Guarding in our safety perspective by bypassing or manipulating post-generation filters, often capitalizing on weaknesses introduced at earlier Input-Level Safety or Training-Time Alignment stages. Such attacks pose significant risks by altering the model's outputs after initial safety checks, necessitating robust guarding mechanisms to intercept harmful content before it reaches the user (Liu et al., 2024; Hu & Chen, 2024).

Hallucination Exploitation These attacks deliberately induce or exploit hallucinations for malicious purposes, leveraging the model's tendency to generate plausible but incorrect information to spread misinformation or bypass safety filters at the inference stage (Liu et al., 2024; Zhu et al., 2025).

- **Adversarial Hallucination Triggering:** Using adversarial prompts or optimization-based strategies to force models to generate false or unsafe information, often evading inference-time guarding mechanisms (Liu et al., 2024).
- **Factual Inconsistency Exploitation:** Leveraging the model's inclination to produce incorrect yet coherent content, particularly dangerous in sensitive domains like healthcare or finance, where hallucinated outputs can appear credible (Wu et al., 2025).
- **Intentional Misinformation Generation:** Crafting prompts specifically designed to produce misleading or fabricated content, undermining trust in model outputs (Liu et al., 2024).
- **Counterfactual Fine-Tuning Risks:** Improper or adversarially manipulated instruction fine-tuning datasets, including counterfactual data constructions, can alter model behavior and influence hallucination patterns (Yu et al., 2024; Sarkar et al., 2025).

Reasoning Manipulation These attacks exploit vulnerabilities in the model's reasoning processes and chain-of-thought (CoT) mechanisms, subverting internal logical pathways to bypass inference-time safety mechanisms and produce unsafe outputs (Yamaguchi et al., 2025; Lin et al., 2025).

- **Chain-of-Thought Jailbreaks:** Structuring prompts or adversarial triggers to manipulate step-by-step reasoning, exploiting CoT generation to increase harmful compliance and attack transferability (Su, 2024; Lin et al., 2025).
- **Internal Reasoning Manipulation:** Intervening in or steering the model's reasoning representations during CoT generation, demonstrating that modifying internal activation patterns can significantly increase unsafe outputs (Yamaguchi et al., 2025).
- **Iterative Reasoning Exploitation:** Injecting adversarial transformations into multi-step reasoning chains to progressively weaken safety alignment and amplify harmful conclusions (Yao et al., 2025; Wu et al., 2025).

Gradient-Based Output Manipulation These attacks use gradient information to manipulate model outputs at the generation level, targeting internal representations to evade Inference-Time Guarding mechanisms designed for text-level safety (Hu & Chen, 2024).

- Gradient Cuff Evasion: Techniques to evade gradient-based jailbreak detection systems by exploiting refusal loss landscapes, bypassing detection mechanisms (Hu & Chen, 2024).
- Embedding Space Attacks: Manipulating prompt embeddings to shift responses from refusal to compliance, operating in the model's internal representation space (Hu & Chen, 2024).
- Activation Steering: Directly manipulating internal activations to control output generation, bypassing traditional safety filters by altering internal representations (Hu & Chen, 2024).

Refusal Circumvention These attacks specifically target the model's refusal mechanisms and safety responses, undermining Inference-Time Guarding by exploiting context-dependent behaviors to elicit unsafe content (Wei et al., 2023; Yu et al., 2023; Ogundoyin et al., 2025; Yu et al., 2024).

- Polite Circumvention: Using overly polite or academic language to bypass refusal triggers, exploiting models' tendency to be more accommodating to polite requests (Wei et al., 2023).
- Explanation-Based Bypasses: Requesting explanations or justifications for harmful content, leading models to provide dangerous information while appearing to maintain safety policies (Yu et al., 2023; Ogundoyin et al., 2025; Yu et al., 2024).
- Hypothetical Framing: Presenting harmful requests as theoretical or educational scenarios, circumventing safety measures by reframing context as non-harmful (Yu et al., 2023; Ogundoyin et al., 2025; Yu et al., 2024).
- Progressive Escalation: Gradually increasing request severity across multiple interactions, exploiting models' context-dependent safety responses to eventually elicit unsafe outputs (Wei et al., 2023).

6.4 Comparative Analysis of LLM Architectures under Adversarial Attacks

While adversarial attacks affect most LLMs, the success and surface of these attacks often depend on the underlying architecture, alignment strategy, and access level of the model. We present a comparative overview of how popular LLM families respond to common attack types in the Table 5.

Discussion: - *GPT-4* demonstrates moderate robustness due to strong alignment layers, but remains vulnerable to jailbreaking and CoT-hijacking. - *Claude* emphasizes ethical alignment via Constitutional AI, often refusing unsafe prompts more reliably (Bai et al., 2022). - *LLaMA-3*, being open-weight, is more prone to fine-tuning-based backdoor insertion and gradient attacks like AutoDAN (Zou et al., 2023).

This comparison highlights that alignment strategies (RLHF vs. Constitutional AI) and openness (API vs. weights) play major roles in attack surface.

Table 5 Comparative overview of major LLMs’ vulnerability to common attacks, highlighting differences in susceptibility across attack types

Attack type	GPT-4 (OpenAI)	Claude 3 (Anthropic)	LLaMA-3 (Meta)
Prompt Injection	Medium	Low [†]	High
Jailbreaks	High	Medium	High
Multimodal Prompt Injection	Supported [‡]	Not Supported	Limited
Backdoor (Fine-tuned)	Low	Unknown	High (in open weights)
Sycophancy Exploitation	Medium	Low	High
Chain-of-Thought Hijack	Medium	Medium	High
Gradient-Based AutoDAN	Low	Not Public	High

[†] Claude uses Constitutional AI for strong refusal tuning.
[‡] GPT-4 Vision supports image inputs, vulnerable to adversarial visual patterns (e.g., FigStep (Gong et al., 2025))

7 Defense Mechanisms

To counteract the adversarial risks, various defense mechanisms (Varshney et al., 2023; Zeng et al., 2024) have been developed, spanning input filtering (Phute et al., 2023), inference-time modifications (Xiong et al., 2024; Ghosal et al., 2024), and alignment techniques (Ghosal et al., 2024; Alami et al., 2024).

In this section, we explore key defense strategies, including rule-based filters (Alon, 2023), inference-time editing, and alignment approaches such as RLHF and Constitutional AI, which are discussed separately in the Safety Alignment section.

Defense strategies against jailbreak attacks on LLMs (Lin et al., 2024) can be categorized into input-stage safeguards and safety-alignment methods. Input-stage safeguards are designed to prevent potential jailbreak prompts from entering the model (for example, rule-based filters). By contrast, defenses that are activated at inference time operate on generated outputs and are aligned with safety-alignment goals; inference-time editing is an example of such a mechanism. These layered defensive controls operate across different stages of the model lifecycle, as illustrated in Fig. 6.

7.1 Rule-Based Filters

Rule-based filters serve as a foundational defense mechanism (Zhang et al., 2023) by preemptively blocking harmful or malicious inputs before they reach the model (Phute et al., 2023). These filters rely on manually curated rules, pattern matching, and keyword-based detection techniques.

- **Keyword Matching:** Detects and blocks prompts containing predefined unsafe terms (Kumar, 2024) e.g., *hate speech*, *violence*
- **Regular Expressions (Regex):** Identifies structured malicious inputs (Siddiq & Zhang, 2024) e.g., prompt injections using “*Ignore all previous instructions*”.
- **Heuristic-Based Detection:** Uses logical conditions to reject queries that follow known attack patterns, such as role-playing attacks or adversarial paraphrasing (Zhang et al., 2023; Liu & Xu, 2024).

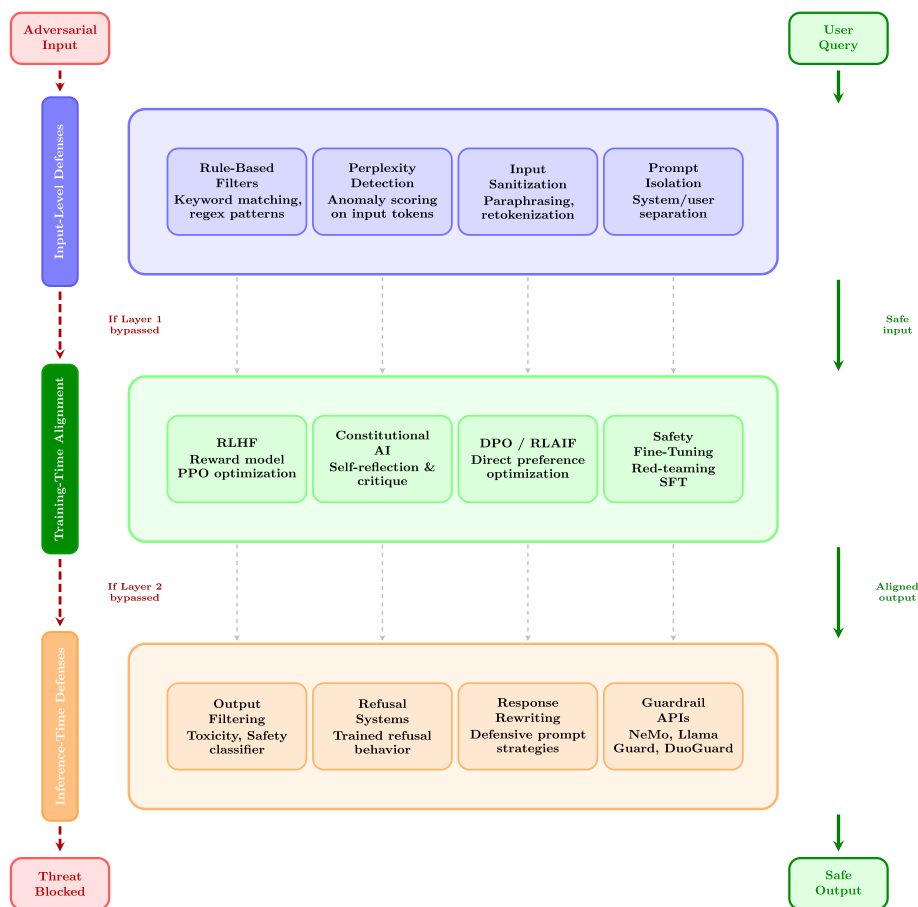


Fig. 6 Layered defense and alignment architecture showing three defense stages. Threats entering from the top-left propagate downward through layers if bypassed (red dashed arrows). Safe queries flow through all layers (green arrows), each adding a level of safety assurance before producing safe output

While effective for well-known threats, rule-based filters struggle against adversarially modified inputs and evolving attack strategies. Therefore, they are often complemented with machine-learning-based defenses.

Limitations and Bypass Cases: Rule-based filters are susceptible to several well-documented bypass strategies. Adversaries can evade keyword matching through synonym substitution, leetspeak encoding, or cross-lingual paraphrasing. Regex-based detection fails against semantically equivalent reformulations that avoid trigger patterns. Heuristic detectors exhibit high false-positive rates on benign role-playing or creative writing prompts, degrading user experience. Known trade-offs include low latency and zero model overhead (cost-efficient) against poor generalization to unseen attack variants and inability to handle context-dependent harm.

7.2 Inference-Time Editing

Inference-time editing refers to modifying the model's generated output (Zhao et al., 2024; Wang et al., 2025) before presenting it to the user. This mechanism ensures that responses align with ethical and safety guidelines without significantly restricting model capabilities.

- **Post-Processing Filters:** Scan generated text for inappropriate content and modify or redact unsafe sections (Song et al., 2024; Wang et al., 2025).
- **Rewriting Mechanisms:** Reformulate responses to remove biases or dangerous implications while preserving the original intent (Xiong et al., 2024).
- **Refusal and Justification Systems:** Instead of outright blocking a response, the model explains why it refuses to answer a potentially unsafe prompt, improving transparency (Wester et al., 2024; Zhang et al., 2025).

These strategies mitigate risks by enforcing real-time corrections while maintaining a balance between safety and usability.

Limitations and Bypass Cases: Post-processing filters can be bypassed when harmful content is embedded within otherwise benign completions or distributed across multiple outputs. Rewriting mechanisms may inadvertently alter factual content or introduce semantic drift, reducing response quality. Refusal systems are vulnerable to explanation-based circumvention, where the model provides harmful details while framing them as justification for refusal. Key trade-offs include: inference-time filters add measurable latency (typically 10 - 100 ms per call depending on classifier size); ensemble-based output moderation increases cost proportionally to the number of classifiers; and overly aggressive post-processing degrades helpfulness on borderline but legitimate queries, risking over-refusal.

8 Safety Alignment

Alignment mechanisms ensure that LLMs generate responses that adhere to ethical, legal, and societal norms.

Unlike traditional defenses, alignment methods shape the model's internal decision-making process rather than filtering inputs or outputs. Two key approaches are Reinforcement Learning from Human Feedback (RLHF) (Wang et al., 2024; Ouyang et al., 2022) and Constitutional AI, as illustrated in Fig. 7.

8.1 Reinforcement Learning from Human Feedback (RLHF)

RLHF (Ouyang et al., 2022) is a widely used method for fine-tuning LLMs based on human preferences and ethical considerations. The process consists of three main steps:

1. **Supervised Pretraining:** The model is fine-tuned using high-quality responses written by human annotators (Liang et al., 2024; Kamath et al., 2024).
2. **Reward Model Training:** Human evaluators rank different model responses, creating a reward function that quantifies response quality (Yang et al., 2024; Li et al., 2024; Wu et al., 2024; Sun, 2024; Wang et al., 2024).

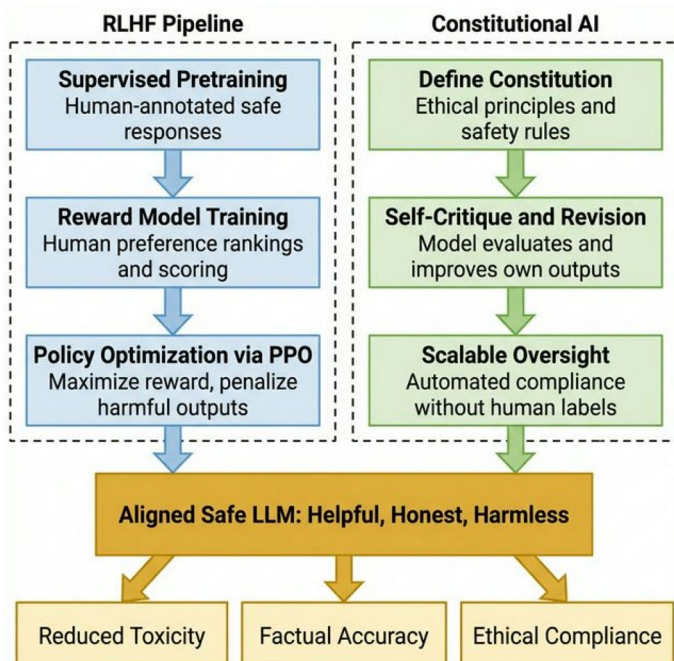


Fig. 7 Overview of safety alignment approaches in LLMs, highlighting the RLHF training pipeline and constitutional AI framework for achieving aligned, safe model behaviour

3. **Policy Optimization:** The model is further fine-tuned using reinforcement learning, optimizing for high-rated responses while penalizing harmful outputs (Wu et al., 2023; Wen et al., 2024).

RLHF has been instrumental in improving LLMs' safety, but its reliance on human feedback introduces biases, and adversarial attacks may still exploit misaligned reward functions.

8.2 Constitutional AI

Constitutional AI is an advanced safety approach (Bai et al., 2022) where models are trained to follow predefined ethical guidelines without requiring extensive human intervention (Henneking, 2025). Instead of learning from human preferences alone, the model optimizes its responses based on a "constitution" of principles (Abiri, 2024).

- **Self-Reflection and Self-Critique:** The model evaluates its own responses and iteratively improves them based on ethical guidelines (Bøe, 2023).
- **Scalability Over RLHF:** Since it does not rely heavily on human-labeled feedback, Constitutional AI can be more scalable and efficient (Dawson, 2024; Kundu et al., 2023; Huang et al., 2024).

By embedding ethical constraints directly into the model's training process, Constitutional AI offers a structured approach to mitigating unsafe behavior at scale.

8.3 Other Alignment Approaches

Beyond RLHF and Constitutional AI, several alternative alignment strategies have emerged that address scalability, annotation cost, and alignment fidelity.

Direct Preference Optimization (DPO): DPO (Rafailov et al., 2023) reformulates the RLHF objective as a supervised classification problem, eliminating the need for an explicit reward model. This reduces training instability and computational overhead while achieving alignment quality comparable to PPO-based RLHF.

Instruction Tuning: Fine-tuning on diverse instruction-following datasets (Wei et al., 2021; Longpre et al., 2023) has proven highly effective for aligning model behavior to human intent without requiring preference annotations. FLAN-style tuning demonstrates that scale and task diversity are key drivers of aligned generalization.

Debate-Based Alignment: Irving et al. (Irving et al., 2018) propose training models through adversarial debate, where two agents argue opposing positions and a human judge selects the winner. This approach is theoretically scalable to superhuman capability regimes where direct human evaluation becomes infeasible.

Red-Teaming as an alignment signal: Automated red-teaming frameworks (Perez et al., 2022; Ganguli et al., 2022) generate diverse adversarial commands to surface model failures, feeding these signals back into alignment training. This closes the loop between attack discovery and safety improvement.

9 Evaluation Framework

9.1 Attack Assessment

The evaluation of attack effectiveness has become increasingly sophisticated, moving beyond simple success rates to comprehensive assessment frameworks. Recent work by (Debenedetti et al., 2024; Chu et al., 2025) introduced refined metrics for measuring attack success (Zhang et al., 2023; Abishethvarman et al., 2025; Downey-Webb et al., 2025), incorporating considerations of transferability and stealth. Their framework demonstrated that attack effectiveness (Zou et al., 2023) must be evaluated across multiple dimensions, including success rate, detectability, and resource requirements.

Semantic preservation (Liu et al., 2023) has emerged as a crucial metric in attack assessment, with recent studies (Wei et al., 2023) showing that many successful attacks achieve their goals at the cost of semantic coherence. This trade-off has important implications for the practical deployment of attack detection systems.

Core Metric Definitions

- **Attack Success Rate (ASR):**

$$\text{ASR} = \frac{\text{Number of successful jailbreaks}}{\text{Total attack attempts}} \quad (1)$$

where success is typically determined by a keyword classifier or an LLM judge (Zou et al., 2023).

- **False Positive Rate (FPR):**

$$\text{FPR} = \frac{\text{Benign prompts incorrectly blocked}}{\text{Total benign prompts}} \quad (2)$$

measuring the over-refusal cost to utility (Jain et al., 2023).

- **Semantic Preservation (BLEU/ROUGE):**

$$\text{BLEU} = \text{n-gram precision with brevity penalty} \quad (3)$$

$$\text{ROUGE} = \text{n-gram recall-based overlap score} \quad (4)$$

These metrics are used to verify that adversarial prompts retain sufficient surface-level coherence to evade detection while achieving attack goals (Zhu et al., 2023).

- **Latency Overhead:**

$$\text{Latency Overhead} = \text{Defense inference time} - \text{Baseline inference time} \quad (5)$$

This metric measures the wall-clock time added per inference call by a defense mechanism, typically reported in milliseconds (Zheng & Rana, 2024).

9.2 Defense Assessment

Defense evaluation has evolved to consider both immediate effectiveness and long-term robustness (Donato, 2025; Carlini et al., 2019). Prior work by (Zhang et al., 2024; Zizzo et al., 2025; Jain et al., 2023) introduced a multi-faceted evaluation framework that considers false positive rates, detection accuracy, and response time. Their research demonstrated the importance of balanced metrics that account for both security effectiveness and practical usability.

Cross-model performance has become increasingly important as organizations deploy multiple LLM variants. Recent studies (Ma et al., 2025) have shown significant variations in defense effectiveness across different model architectures, highlighting the need for more generalizable defense strategies.

9.3 Towards a Standardized Evaluation Protocol

A persistent obstacle to reproducible LLM safety research is the absence of a shared evaluation protocol. Across the papers surveyed, attack success rate (ASR) is computed with at least four different success criteria (keyword match, LLM judge, human annotation, and regex refusal detection), making cross-paper comparison unreliable. We propose the following minimum reporting standard for future work.

Adopting Table 6 as a community standard would allow meta-analyses across attack and defense papers, enable fair comparison of guardrail systems, and reduce the risk of inflated ASR claims caused by inconsistent judge selection.

10 Security Measures

Safety mechanisms at the inference stage (Welbl et al., 2021) play a critical role in preventing harmful or undesirable outputs (Gehman et al., 2020) from reaching the end user. After a language model generates a response, these guarding mechanisms intercept and evaluate the content to detect toxicity (Goyal et al., 2024; Wang et al., 2024), misinformation (Liu et al., 2024), bias (Ayyamperumal & Ge, 2024), or policy violations (O'Neill et al., 2024). Common techniques include:

- **Output Filtering:** Automatically screening generated text for offensive language, hate speech, coded toxicity, or disallowed content using classifier-based or knowledge-augmented detection systems, blocking or sanitizing it before delivery (Zhao et al., 2025; Hackmann, 2024; Daquigan et al., 2025).
- **Toxicity and Bias Detection:** Using specialized deep learning classifiers, fairness-aware models, or prompt-based self-diagnosis mechanisms to assess whether outputs contain harmful biases, implicit toxicity, or group-targeted discrimination (Mostafazadeh Davani et al., 2021; Chen et al., 2024; Wang & Chang, 2022).
- **Prompt and Output Sanitization:** Removing, masking, or reformulating sensitive, misleading, or harmful segments of the response—potentially leveraging contextual analysis or linguistic signal extraction, while preserving the overall intent (Kikkiseti et al., 2024; Han et al., 2020; Wang et al., 2026; Mo et al., 2025).
- **Rule-based or Learned Guardrails:** Incorporating hard-coded safety constraints, configurable policy frameworks, reasoning-based guard models, or multi-layered runtime architectures that enforce policy-aligned behavior, detect violations across domains, and

Table 6 Proposed minimum reporting dimensions for LLM safety evaluations

Dimension	Required specification	Example values
Model access	Black-box/White-box/Gray-box	Query-only API; full weights
Attack surface	Prompt-only/RAG+Tool/Training-time	System prompt; retrieval store
Attack budget	Queries, turns, or optimization steps	500 queries; 20 GCG steps
Policy scope	Disallowed content categories in scope	Violence, CSAM, CBRN
Success criterion	ASR definition and judge type	Keyword classifier; GPT-4 judge
Defense overhead	Latency and throughput delta	+12 ms/call; −8% throughput
Utility under benign traffic	Task accuracy on clean prompts	MT-Bench score; MMLU delta

dynamically adapt to evolving risk contexts (Wu et al., 2025; Rad et al., 2025; Sreedhar et al., 2025; Wang & Li, 2025; Avinash et al., 2025; Li et al., 2025; Wen et al., 2025).

- **Human-in-the-Loop Review:** In high-risk or ambiguous scenarios, routing generated responses for human moderation (Crisan & Fiore-Gartland, 2021), potentially supported by explanation-generating models to assist decision-making before release (Yadav et al., 2024; Alvisi et al., 2025).

These inference-time safety measures act as a final checkpoint, complementing earlier training-time alignment and input-level defenses. By monitoring, steering, and controlling the model's outputs in real-time, they help ensure safer interactions while maintaining user trust and system reliability.

10.1 Guardrails

A guardrail (Dong et al., 2024a) is an algorithmic or procedural safeguard that takes the model's input and/or output and enforces safety policies through real-time intervention. When a user's prompt or the resulting model output raises safety concerns (e.g., relating to violence, exploitation, or misinformation), guardrails may suppress the response or adapt it to ensure harmlessness and compliance with ethical standards (Dong et al., 2024b; Yang et al., 2024; Ravichandran et al., 2025).

Guardrail mechanisms can be situated at several intervention points:

- **Input/Query Filtering:** Identifying misuse or policy-violating intent at the query stage to block unsafe requests before generation begins (Akheel, 2025; Chua et al., 2025; Dong et al., 2024; Hasan et al., 2025; Li et al., 2025; Zhuang et al., 2025)
- **Post-Generation Output Moderation:** Screening and redacting model outputs based on toxicity, bias, or ethical violations, using classifiers, regular expressions, or human-in-the-loop review systems (Banerjee et al., 2024; Deng et al., 2025; Irtiza et al., 2024; Rebedea et al., 2023).
- **Dynamic Behavioral Constraints:** Enforcing context-sensitive policies or dynamic constraints at runtime to adapt model behavior, as with on-the-fly filtering and response reshaping (Abdelkader et al., 2024).

Recent research has expanded the sophistication and deployment points of guardrails:

Pre-Training Guardrails: Focus on building safe and diverse datasets to prevent the model from encoding harmful content from the outset (O'Neill et al., 2024; Zheng et al., 2024).

Training-Time Interventions: Use of bias mitigation (Dada, 2024; Nagireddy et al., 2024) and fairness constraints (Pantha et al., 2024) to proactively reduce the model's intrinsic incentives toward unsafe behaviors.

Post-Training and Deployment Filtering: Incorporate advanced output moderation systems, ensemble classifiers, or modular tools (like Guardformer, LLMGuard) to handle safety at scale in live applications (Deng et al., 2025; O'Neill et al., 2024).

Table 7 Overview of major LLM guardrail studies, highlighting their intervention stage, methodology, evaluation, and notable outcomes

Paper	Stage [†]	Technique	Eval dataset	Key result
Building Guardrails for Large Language Models (Dong et al., 2024)	IF/OF	Neural-symbolic rules	Case studies	0 critical policy leaks
A Flexible Large Language Models Guardrail Development Methodology Applied to Off-Topic Prompt Detection (Chua et al., 2025)	IF	Synthetic data generation	10K off-topic prompts	−21% FPR
GuardBench: A Large-Scale Benchmark for Guardrail Models (Bassani & Sanchez, 2024)	–	Benchmark suite	GuardBench (40 sets)	Std. scoring API
Swiss Cheese Model for AI Safety: A Taxonomy and Reference Architecture for Multi-Layered Guardrails of Foundation Model Based Agents (Shamsujjoha et al., 2025)	RT	Design taxonomy	Theoretical	Latency–coverage trade-off
Self-Evaluation as a Defense Against Adversarial Attacks on LLMs (Brown et al., 2024)	OF	Self-Evaluation	SafetyBench	−41% Jailbreak ASR
DuoGuard: A Two-Player RL-Driven Framework for Multilingual LLM Guardrails (Deng et al., 2025)	OF	Dual-agent RL	9-lang toxic set	−33% Toxicity
NeMo Guardrails: A Toolkit for Controllable and Safe LLM Applications with Programmable Rails (Rebedea et al., 2023)	IF/OF	YAML rule engine	RealToxicity	0.92 F1 vs GPT judge
Lightweight Safety Guardrails Using Fine-tuned BERT Embeddings (Zheng & Rana, 2024)	IF	BERT detector	30K prompts	75× speed-up
Distilling On-device Language Models for Robot Planning with Minimal Human Intervention (Ravichandran et al., 2025)	OF	Robot-aware policy	Sim-Nav env	98% safe plans

[†] IF = Input Filtering, OF = Output Filtering, RT = Runtime architectural guidance

10.2 Recent Advances in Guardrail Research

The last eighteen months have produced a rapid expansion of dedicated guardrail techniques that complement alignment and adversarial training that is listed in the Table 7. (Dong et al., 2024) introduce a socio-technical, neural-symbolic design flow that starts from policy elicitation and ends with formally verified guardrail code. (Chua et al., 2025) present a data-free methodology that synthesises a large prompt corpus to bootstrap off-topic detectors; their guardrail reduces false positives by 21% compared with heuristics. Prior work such as (Bassani & Sanchez, 2024) tackle the evaluation gap with GuardBench, a 40-dataset benchmark and an automated Python pipeline that standardises guardrail metrics across languages and domains. Beyond input filtering, runtime taxonomies have emerged. (Shamsujjoha et al., 2025) maps the guardrails along three architectural axes *motivation*, *quality attributes*, *design options* and shows how these choices affect latency and coverage. (Brown et al., 2024) proposes Self-Guard, a light-weight self-reflection layer that detects policy violations in the model’s own draft answer before final emission, cutting jailbreak ASR by 41% on SafetyBench. (Deng et al., 2025) extend the concept to multilingual settings with DuoGuard, a two-agent RL framework that negotiates safe responses across nine languages without sacrificing helpfulness. Toolkits are maturing as well. For instance, Rebedea et al. (Rebedea et al., 2023) release NeMo Guardrails, a programmable rule engine that supports

regex, semantic embeddings and LLM-judge policies inside a single YAML specification. Zheng and Rana (2024) show that fine-tuning a compact BERT detector on only 30k safety prompts yields a 75 times faster guardrail while matching GPT-based moderation accuracy. Finally, Ravichandran et al. (2025) explore embodied scenarios: their Safety Guardrails for Robots integrate motion-planning constraints with language filters, demonstrating that policy leaks in the language channel can propagate to unsafe trajectories.

Guardrail research is converging on three complementary trends: (i) *holistic benchmarking*, (ii) *instruction-level pre-training*, and (iii) *low-latency orchestration*.

Holistic Benchmarking. Prior work such as Wang et al. (2025) present the first Systematization-of-Knowledge dedicated to jailbreak defenses; their SEU framework quantifies trade-offs between security, efficiency and utility across 18 public rails, revealing that even the best guards still allow 29% median ASR.

Instruction-Level Pre-training. Guardformer proposes “guardrail instruction pre-training” that injects 1.2M safety prompts during SFT, cutting average jail-breakability by 43% while adding only 0.9% training cost (O’Neill et al., 2024). Complementarily, LLM-GUARD couples a lightweight classifier with a self-reflection prompt and achieves 0.92 F_1 on RealToxicity while introducing < 5 ms latency per call (Goyal et al., 2024).

Low-Latency Orchestration. “When-in-Doubt, Cascade” shows that a two-stage pipeline (fast BERT gate $\rightarrow$ large LLM judge) preserves 98% task utility yet halves cost relative to LLM-only moderation (Nagireddy et al., 2025). At the 1 level, Ayyamperumal and Ge survey the risks associated with deploying LLMs and outline a layered protection model for implementing guardrails across external, secondary, and internal levels to mitigate unsafe behaviour and reliability concerns (Ayyamperumal & Ge, 2024)

11 Future Directions

This survey identifies several open research directions across the five pillars of LLM safety:

Multimodal Safety: Current guardrail and alignment techniques are primarily designed for text. Multimodal LLMs introduce new attack surfaces including adversarial visual patterns, audio manipulation, and cross-modal prompt injection (Gong et al., 2025; Qi et al., 2023). Dedicated multimodal safety benchmarks and cross-modal guardrails remain an open problem.

Agentic and Multi-agent Safety: As LLMs are increasingly deployed as autonomous agents with tool access and memory, new threat vectors emerge including tool-call injection, memory poisoning, and cascading failures across agent pipelines (Shayegani et al., 2023; Wang et al., 2025). Safety frameworks must evolve to address these agentic settings where a single compromised agent can propagate harm across an entire system.

Scalable Alignment: RLHF and Constitutional AI face scalability bottlenecks as models grow larger and task complexity increases. Future work should explore scalable oversight mechanisms (Irving et al., 2018), debate-based alignment, and weak-to-strong generalization frameworks that maintain alignment guarantees beyond human-evaluable capability thresholds.

Formal Verification of Guardrails: Current guardrail evaluation is largely empirical. Formal methods for verifying safety properties of guardrail systems – analogous to program

verification – remain nascent. Developing provable safety bounds for guardrail coverage and false-negative rates is a critical open challenge (Dong et al., 2024).

Cross-Lingual and Low-Resource Safety: Most safety benchmarks and alignment datasets are English-centric. Models deployed in multilingual settings exhibit substantially degraded safety performance in low-resource languages, as attackers can exploit this gap through crosslingual bypass (Deng et al., 2025). Building safety-aware multilingual benchmarks and transfer-learning approaches for alignment is an underexplored priority.

12 Limitations

This survey provides a structured overview of LLM safety, focusing on attacks, defenses, alignment, evaluation metrics, and guardrails. However, it has certain limitations. We cover only specific attack types rather than an extensive list of all possible threats. Likewise, our discussion on defenses is limited to countermeasures for these selected attacks, excluding broader techniques like adversarial training or differential privacy. Our evaluation metrics are also narrowly scoped, considering only attack and defense assessments while leaving out aspects like fairness, user experience, and ethical compliance. Additionally, our safety discussion is restricted to guardrails, without addressing areas like interpretability, uncertainty estimation, or adversarial robustness in detail. Given the rapid pace of advancements in LLM research, some techniques and findings discussed in this survey may become outdated over time, necessitating continuous updates to maintain relevance. These limitations highlight the need for future research to explore a wider range of attack types, stronger defense mechanisms, alternative alignment strategies, and broader safety measures beyond guardrails.

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Declarations

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